1. What is CRUD in FastAPI?
   CRUD stands for Create, Read, Update, and Delete, these are the four basic operations we perform on data. In FastAPI, we map these to HTTP methods like POST, GET, PUT, and DELETE to build a complete API.

2. Why do we use the POST method for inserting data?
   POST is designed to send new data to the server, which is why it's used for insertion. Unlike GET, it carries data in the request body, making it secure and suitable for creating new records.

3. Difference between PUT and PATCH
   PUT replaces the entire resource with new data, while PATCH only updates the specific fields you provide. Consider PUT as "replace everything" and PATCH as "just fix what I mention."

4. What is the path parameter?
   A path parameter is a variable part of the URL, like /students/{id}, where {id} changes based on the student you want. FastAPI automatically extracts and adds it into your function.

5. What is the request body in FastAPI?
   The request body is the data sent by the client in a POST or PUT request, usually in JSON format. FastAPI reads it and maps it directly to your Pydantic model automatically.

6. Why do we use Pydantic models?
   Pydantic models help us validate and structure the incoming data automatically. If someone sends wrong data types, FastAPI throws a clean error before it even reaches your logic, no manual checks needed.

7. What status code is returned for successful deletion?
   Typically, 204 No Content is returned after a successful deletion, meaning the action was done but there's nothing to send back. Sometimes 200 OK is also used if you return a confirmation message.

8. How can we check whether a student exists before updating?
   We can look up the ID in the data store (like a dictionary or list) before updating. If it's not found, we raise an HTTPException with a 404 Not Found status to inform the client.

9. What happens if duplicate IDs are inserted?
   If you're using a dictionary, the old record gets silently overwritten by the new one. To prevent this, you should check if the ID already exists and raise a 400 Bad Request error if it does.

10. How does FastAPI automatically generate Swagger UI?

FastAPI uses the OpenAPI standard and auto-reads your routes, models, and type hints to build the docs. Just visit /docs in the browser and we will get a fully interactive Swagger UI, no extra setup needed.